

Ila Rehman

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Summary

MCA graduate and self-taught AI engineer with a physics background, focused on shipping LLM-powered applications. Built a multi-tenant AI sales agent (**LangGraph**, **FastAPI**) and a production **RAG** API (**FastAPI**, **LangChain**, **FAISS**, **GPT-4o**), and currently build distributed, event-driven **AWS** microservices with **LLMs** integrated into production pipelines as a software engineer intern.

Experience

Manrke

April 2026 – Present

Software Engineer Intern

Remote

- Design and build distributed, event-driven microservices on **AWS** (**Lambda**, **ECS**, **Step Functions**, **SQS/SNS**, **S3**, **DynamoDB**, **RDS**) powering an end-to-end document-automation platform.
- Own systems across the full lifecycle: architecture, implementation, containerized deployment to **ECR/Lambda** across multiple **AWS** accounts, monitoring, and live production debugging.
- Integrate **LLMs** into production pipelines for extraction, classification, and natural-language generation, with guardrails, deterministic fallbacks, and cost/latency controls.
- Engineer for reliability at scale: **idempotent processing**, **exactly-once semantics**, distributed state coordination, and graceful failure handling across asynchronous services.
- Drive observability and operations: structured logging, correlation-ID tracing through **CloudWatch**, and alerting that surfaces and diagnoses issues quickly.

Projects

AI SDR (Davis) [\[code\]](#) [\[live demo\]](#)

Next.js, **React**, **TypeScript**, **FastAPI**, **LangGraph**, **Groq**, **Supabase**, **Pinecone**, **Redis**

- Built an evidence-grounded AI sales-development agent that finds prospects, researches them, and drafts personalized outreach, where every prospect score cites a source URL, snippet, and observed-at timestamp.
- Split scoring into an **LLM** signal-extraction stage and a deterministic **Python** scorer with weighted ICP criteria, so identical inputs always produce identical, unit-testable scores; added an approval-first send pipeline (pending, approved, sent) with a 6-rule policy guard that makes duplicate sends impossible via a unique idempotency key.
- Shipped multi-tenant isolation across organizations and workspaces enforced by **row-level security** over 45 **Postgres** tables, with **Supabase** auth, per-workspace **Pinecone** namespaces, and **Braintrust** tracing; deployed frontend on **Vercel** and backend on **Railway**.

AdaptQuiz API [\[code\]](#) [\[live demo\]](#)

Python, **FastAPI**, **LangChain**, **FAISS**, **GPT-4o**, **Docker**, **Railway**

- Built and deployed a production **RAG** (Retrieval-Augmented Generation) API that turns uploaded PDF/TXT study material into adaptive quizzes grounded in the source text, removing hallucinated questions.
- Built an **LLM-as-judge** grader (**GPT-4o**) scoring answers on a 3-criterion rubric (accuracy, completeness, terminology) with partial credit and per-question knowledge-gap tags, returned as structured JSON; served over 6 **REST** endpoints (**OpenAPI**/Swagger, **Pydantic** v2), containerized with **Docker** on **Railway**.

Technical Skills

Languages: Python, SQL, JavaScript / TypeScript

AI & LLM: LLM APIs (OpenAI GPT-4o, Groq, Anthropic Claude), RAG (Retrieval-Augmented Generation), LangChain, LangGraph, AI Agents, Prompt Engineering, Structured Outputs, Tool / Function Calling, LLM-as-Judge Evaluation, Embeddings (sentence-transformers), Semantic Search, FAISS, Pinecone

Backend: FastAPI, REST API Design, Pydantic, SQLAlchemy / Alembic, PostgreSQL, Supabase, DynamoDB, Redis, Event-Driven Microservices, Multi-Tenant Architecture, Row-Level Security, Next.js

Cloud & DevOps: AWS (Lambda, ECS, Step Functions, SQS/SNS, S3, DynamoDB, RDS, ECR, CloudWatch), Docker, Railway, Vercel, Git & GitHub, CI/CD

Tools & Foundations: pytest, Braintrust, OpenAPI / Swagger, NumPy, Pandas, ML Evaluation Metrics, Linear Algebra, Statistics

Education

Master of Computer Applications (MCA)

2022 – 2025

Indira Gandhi National Open University (IGNOU), New Delhi

Coursework: Software Engineering, Data Science, Cloud Computing, Artificial Intelligence & Machine Learning

B.Sc. (Hons) Physics with Mathematics

2017 – 2020

Jamia Millia Islamia, New Delhi

First Class Honours